

IZHEVSK, Russian Federation

WMO: 284110

Lat: 56.833N

Lon: 53.450E

Elev: 512

StdP: 14.43

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.2	-14.1	-25.8	1.3	-19.6	-19.3	1.9	-13.8	24.0	20.7	21.7	20.5	3.7	340	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	85.6	68.0	82.3	66.4	79.1	65.0	70.0	81.4	68.4	79.4	66.6	76.5	6.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	98.4	75.1	64.5	92.5	73.5	62.7	86.9	71.7	34.2	81.0	32.9	79.6	31.5	76.9	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	21.1	17.9	15.5	DB	-26.0	89.7	6.8	3.4	-30.9	92.2	-34.8	94.2	-38.7	96.1	-43.6	98.5
(5)				WB	-26.2	72.9	6.8	2.3	-31.0	74.6	-35.0	76.0	-38.8	77.3	-43.7	79.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	9.3	11.3	23.6	39.3	53.8	61.8	65.8	61.8	51.3	38.6	24.6	13.5	(6)
(7)		DBStd	22.39	12.97	12.43	8.06	9.49	9.32	7.99	6.94	8.10	7.89	8.27	11.22	12.97	(7)
(8)		HDD50	5925	1263	1084	818	345	70	8	0	5	75	364	762	1131	(8)
(9)		HDD65	10079	1728	1504	1283	772	362	153	76	162	414	817	1212	1596	(9)
(10)		CDD50	1559	0	0	0	23	188	361	490	370	115	12	0	0	(10)
(11)		CDD65	238	0	0	0	0	14	56	101	62	5	0	0	0	(11)
(12)		CDH74	2154	0	0	0	2	172	504	903	539	34	0	0	0	(12)
(13)		CDH80	573	0	0	0	0	29	118	247	175	4	0	0	0	(13)
(14)	Wind	WSAvg	7.5	7.7	7.8	8.4	7.8	8.0	6.9	5.7	6.3	6.7	8.1	8.2	7.8	(14)
(15)	Precipitation	PrecAvg	20.7	1.5	1.1	1.1	1.2	1.3	2.3	2.7	2.4	2.0	2.0	1.8	1.5	(15)
(16)		PrecMax	28.2	2.7	2.9	3.8	2.5	3.6	4.6	4.6	9.3	4.8	3.8	3.5	3.1	(16)
(17)		PrecMin	14.1	0.3	0.1	0.1	0.4	0.4	0.2	0.6	0.5	0.5	0.9	0.4	0.1	(17)
(18)		PrecStd	3.6	0.7	0.6	0.8	0.6	0.8	1.1	1.1	1.8	1.1	0.8	0.8	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	34.9	43.8	72.5	83.6	88.0	89.9	90.5	78.8	64.7	48.2	36.2	(19)
(20)		MCWB	32.5	32.8	37.6	56.3	63.0	68.2	68.4	67.6	61.1	53.0	46.1	34.9		(20)
(21)		2%	DB	32.3	33.2	39.5	65.7	78.8	83.3	85.7	84.8	72.5	58.4	43.5	34.0	(21)
(22)		MCWB	31.3	31.9	34.8	52.7	60.6	66.6	68.8	67.6	59.6	50.2	41.7	32.8		(22)
(23)		5%	DB	30.2	31.6	36.6	59.2	74.5	79.9	82.4	79.8	67.7	53.7	39.9	32.6	(23)
(24)		MCWB	28.9	30.5	33.2	48.9	58.6	65.2	67.4	66.1	57.5	48.2	38.3	31.7		(24)
(25)		10%	DB	27.3	28.5	34.5	53.7	70.3	76.2	79.2	75.3	63.8	50.2	36.1	30.3	(25)
(26)		MCWB	26.1	27.2	32.1	45.8	56.6	63.4	66.2	64.3	56.0	46.6	34.7	29.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	33.5	38.2	57.1	66.2	71.1	72.7	71.0	63.7	55.1	46.5	35.1	(27)
(28)		MDCB	33.8	34.0	42.4	70.5	78.3	82.3	83.5	82.7	73.1	60.6	47.9	35.7		(28)
(29)		2%	WB	31.4	32.2	35.6	53.3	62.5	68.4	70.4	69.1	60.9	51.5	42.0	33.1	(29)
(30)		MDCB	32.2	33.0	38.3	64.2	75.6	80.2	81.4	81.3	69.4	56.8	43.4	33.8		(30)
(31)		5%	WB	29.1	30.5	33.9	50.0	60.1	66.6	69.0	67.0	58.7	49.1	38.5	31.7	(31)
(32)		MDCB	30.1	31.7	35.8	57.7	71.6	77.3	79.6	78.0	65.9	52.9	39.8	32.5		(32)
(33)		10%	WB	26.2	27.4	32.4	46.2	57.5	64.6	67.4	64.9	56.7	46.8	35.0	29.3	(33)
(34)		MDCB	27.2	28.6	34.3	53.5	68.0	74.1	76.9	73.7	63.1	50.1	35.9	30.3		(34)
(35)	Mean Daily Temperature Range	MDBR	10.9	12.2	12.8	15.0	19.1	17.9	18.3	16.9	14.8	10.0	7.8	9.6	(35)	
(36)		5% DB	MCDBR	9.3	9.0	13.1	23.0	24.9	22.7	23.1	22.7	21.3	16.7	8.5	7.8	(36)
(37)		MCWBR	9.3	8.2	9.8	13.0	12.0	10.0	9.7	9.4	11.0	10.4	7.8	7.6	(37)	
(38)		5% WB	MCDBR	9.3	8.5	10.6	21.6	22.8	20.2	20.6	20.7	19.0	14.9	8.0	7.2	(38)
(39)		MCWBR	9.5	8.3	8.8	13.3	11.9	10.1	9.9	9.8	10.7	10.2	7.8	7.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.291	0.316	0.371	0.402	0.389	0.385	0.407	0.425	0.372	0.350	0.310	0.286	(40)	
(41)		taud	2.026	2.029	1.950	2.098	2.305	2.367	2.314	2.237	2.357	2.285	2.141	2.003	(41)	
(42)		Ebn at Noon	173	223	239	255	269	272	262	246	245	213	167	137	(42)	
(43)		Edh at Noon	20	30	43	43	37	36	37	37	29	24	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	198	495	936	1353	1692	1816	1799	1363	841	373	155	110	(44)	
(45)		RadStd	29	71	96	145	167	151	175	127	101	67	30	26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page